# Statistics

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- Statistics
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# Statistics

The practice of collecting, analysing, interpreting and communicating data. It id the science of interacting with data.

Types of Statistics

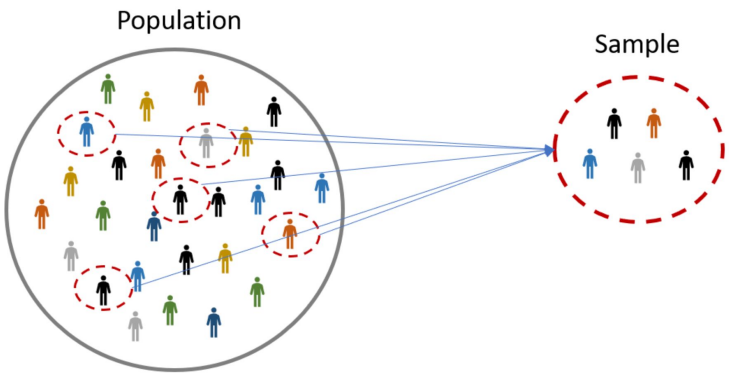
1. **Descriptive** statistics is getting the data and talking about it.
2. **Inferential** statistics deals with taking the sample and analysing the sample to make judgement and claim about it.

**The point is that statistics helps us process data, which is really just a fancy name for information.**

# Population & Sample

**Populations** refers to the total amount of things, where as **sample** refers to a small part of the population that is used for study.

**Sample size** is the total amount of things in a sample.



# Dataset

**Data items** is characteristic of a data unit

Data unit is one entity in the population being studied, about which data are collected.

| | age | sex | income |
|---|---|---|---|
| John Smith | 18 | M | 50K |
| Joe Bloggs | 18 | M | 40K |
| Sally Jones | 20 | F | 55k |
| Linda Lee | 22 | F | 50K |
| Harry James | 19 | M | 35k |

**Datum or Observation** is an occurrence of a specific data item that is recorded about a data unit.

# Dataset

**Data** are measurement or observation that are collected as a source of information. So, we can say data is merely the raw material of knowledge.

Data is further classified in two types.

1. Quantitative or measurement of value or counts.
2. Qualitative or measure of type.

# Dataset

**Data items or Variable** is a characteristic of a data unit that can be measured or counted. Data items further divided into two types.

1. **Numerical** variables have values that describe a measure quantity as a numbers.
    a. **Continuous** observations is a set of real numbers.
    b. **Discrete** observations is a set of distinct whole number.
2. **Categorical** variable have value that describe a quality of a data unit.
    a. **Ordinal** observations that can be logically ordered or ranked.
    b. **Nominal** observations that can not be logically ordered or ranked.

# Frequency

The **frequency** is the number of times a particular value of a variable(data item) has been observed to occur. We can describe the frequency in two types.

1. **Absolute frequency** describes the number of times a particular value of a variable has been observed to occur.
2. **Relative frequency** describes the number of times a particular values of a variable has been observed to occur in relation to the total number of value for that variable.

Ratio, Rate, Proportion and Percentage used for expressing relative frequency.

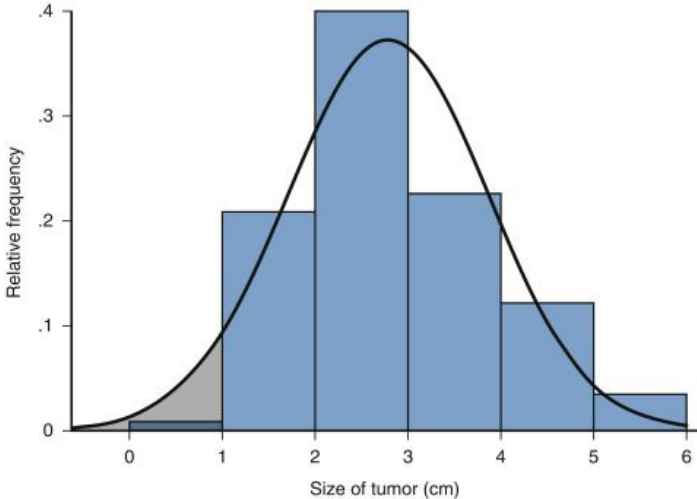
# Frequency

Suppose tax on Car purchase is raised from 3% to 5%. There are two ways to express this tax change, both of which are technical accurate. The government, who engineered this tax increase, pointed out (correctly) that the tax on purchasing car was increased by 2%(from 3% to 5%). The opposition pointed out(also correctly) that the tax on purchasing car has been raised by 67%[ (5-3)/3 ].

The government is focused on the absolute change in the tax rate; the opposition focussed on the % change in the tax burden. As noted, both descriptions are technically correct, though I would argue that the opposition description more accurately conveys the impact of the tax change, since what I'm going to have to pay to the government - The amount that I care about, as opposed to the way it is calculated- really has gone up by 67%.

# Frequency

**Frequency distribution** are visual distribution that organise and present frequency count so that the information can be interpreted more easily.



# Measurement of Central tendency

**Measurement of central tendency** is a summary measure that attempts to describe a whole set of data with a single value that represents the middle or centre of its distribution. So, we can describe the measurement of central tendency into three types.

1. **Mode** is the most common occurring value in distribution.
2. **Median** is the middle value in distribution when the values are arranged in ascending or descending order.
3. **Mean** is the sum of the value of each observation in a dataset divided by the number of the observations.

# Measurement of Central tendency

Consider this dataset showing retirement age of 11 people in a whole years.

54, 54, 54, 55, 56, 57, 57, 58, 58, 60, 60

**Mode : ? => 54**

**Median: ? => 57**

**Mean: ? => (623/11) = 56.6**

| Age | Frequency |
|---|---|
| 54 | 3 |
| 55 | 1 |
| 56 | 1 |
| 57 | 2 |
| 58 | 2 |
| 60 | 2 |

# Measurement of Central tendency

**The mode** has an advantage over the median and the mean as it can be found for both numerical and categorical data.

There are some limitation to using mode.

- In some cases, like previous example the mode may not reflect the centre of the distribution.
- It is also possible for there to be more than one mode for the same distribution.
- In some cases, particularly where the data are continuous, the distribution may have no mode at all(i.e. All values are different).

In cases such as these, it may be better to consider using the median or mean or group the data into appropriate intervals and find modal class.

# Measurement of Central tendency

The **median** is less affected by outliers and skewed data than the mean and is usually the preferred measure of central tendency when the distribution is not symmetrical.

The **median** can not be identified for categorical nominal data, as it can't be logically ordered.

The **mean** can be used for both continuous and discrete numeric data.

The **mean** can't be calculated for categorical data, as the values can't be summed. As the **mean** includes every value in the distribution the mean is influenced by outliers and skewed distributions.

# Measurement of Central tendency

Imagine that ten guys sitting on bar stools in a middle-class drinking bar; each if these guys earns $35000 a year, which makes the mean annual income for the group $35000. Bill gates walk into the bar. Let's assume for the sake of example that Bill Gates has an annual income of $1 billion. When Bill sits down on the eleventh bar stool, the mean annual income for the bar patrons rises to about $91 million. Obviously none of the original ten drinkers is any richer. If I were to describe the patrons if this bar as having an average annual income of $91 billion, the statement would be both statistically cross and gross misleading. This isn't bar where multi millionaires hang out; it's a bar where a bunch of guys with relatively low incomes happen to be sitting next to Bill Gates. **THE SENSITIVITY OF MEAN TO OUTLIERS.**

# Measurement of Central tendency

- Median is physical middle point.
- Mean can be thought as balance point.

**Population Mean**

$$\mu = \frac{\sum x}{N}$$

**Sample Mean**

$$\bar{X} = \frac{\sum x}{n}$$

# Measurement of Spread

Measurement of spread describe how similar or varied the set of observed values are for a particular variable. Measure of spread includes

- **Range** is the difference between the smallest and the largest value in a dataset.
- **Quartile** divide an ordered dataset into four equal parts.
- **Interquartile range** is often seen as a better measure of spread than the range as it is not affected by outliers.

  IQR = Q3 - Q1

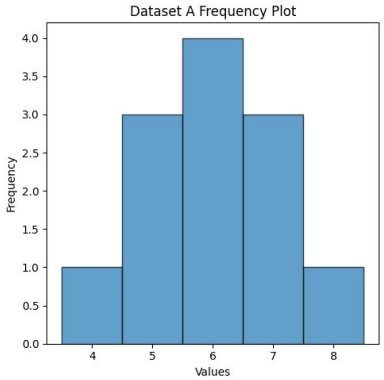
- **Variance & Standard deviation** summarise how close each observed data value is to the mean value.

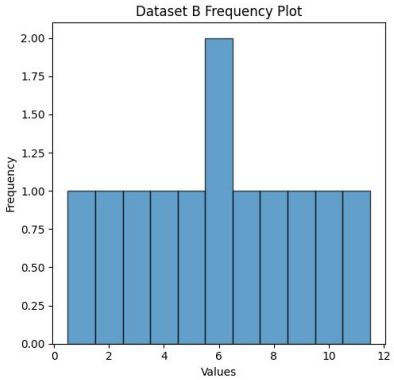
# Measurement of Spread

Summarising the dataset can help us to understand the data, especially when the dataset is large. As discussed in the measurement of central tendency, the mode, median and mean summarise the data into a single value that is typically or representative of all the values in the dataset but this is only the part of the picture that summarises a dataset. Measure of spread summarise the data in away that shows how scattered the values are and how much they differ from the mean value.

# Measurement of Spread

| Dataset A | Dataset B |
|---|---|
| 4, 5, 5, 5, 6, 6, 6, 6,7,7,7,8 | 1,2,3,4,5,6,6,7,8,9,10,11 |
| Mode:6, Mean: 6, Median: 6 | Mode:6, Mean: 6, Median: 6 |




If we just looked at the measurement of central tendency, we may assume that the datasets are the same.

However, if we look spread of the values in the following graph, we can see that Dataset B is more depressed that Dataset A. Used together, the measures of spread helps us to better understand the data.

# Measurement of Spread

| Dataset A | Dataset B |
|---|---|
| 4, 5, 5, 5, 6, 6, 6, 6,7,7,7,8 | 1,2,3,4,5,6,6,7,8,9,10,11 |
| Mode:6, Mean: 6, Median: 6 | Mode:6, Mean: 6, Median: 6 |
| Number Line view: 4-5-6-7-8 | Number Line view: 1-2-3-4-5-6-7-8-9-10-11 |

Range is the difference between the smallest value and the largest value in a dataset.

Range A: 8 - 4 = 4

Range B: 11 -1 = 10

We can say Dataset B is more spread than Dataset A.

# Measurement of Spread

| Dataset A | Dataset B |
|---|---|
| 4, 5, 5, 5, 6, 6, 6, 6,7,7,7,8 | 1,2,3,4,5,6,6,7,8,9,10,11 |
| Mode:6, Mean: 6, Median: 6 | Mode:6, Mean: 6, Median: 6 |
| 4, 5, 5, Q1, 5, 6, 6, Q2, 6, 6,7, Q3, 7,7,8 | 1,2,3, Q1, 4,5,6, Q2, 6,7,8, Q3, 9,10,11 |
| Q1 = (5+5)/2 =5<br>Q2 = (6+6)/2 = 6<br>Q3 = (7+7)/2 = 7 | Q1 = (3+4)/2 =3.5<br>Q2 = (6+6)/2 = 6<br>Q3 = (8+9)/2 = 8.5 |
| IQR = Q3 - Q1 = 7 - 5 = 2 | IQR = Q3 - Q1 = 8.5 - 3.5 = 5 |

# Measurement of Spread

The variance and standard deviation are measures of the spread of the data around the mean. They summarises how close each observed data value is to the mean value.

The smaller the variance and standard deviation, the more the mean value is indicative of the whole data set. If all values of a dataset are the same the standard deviation and variance are zero.

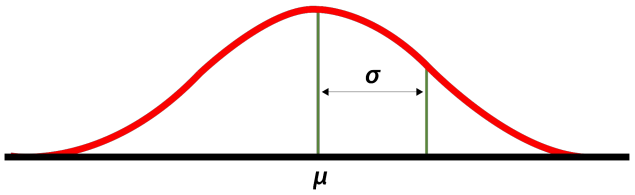
**Variance and Standard Deviation Formula**

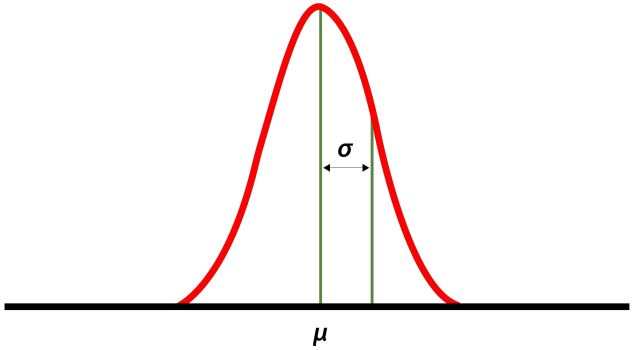
| | Population | Sample |
|---|---|---|
| Variance | $\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$ | $S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$ |
| Standard Deviation | $\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$ | $S = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$ |

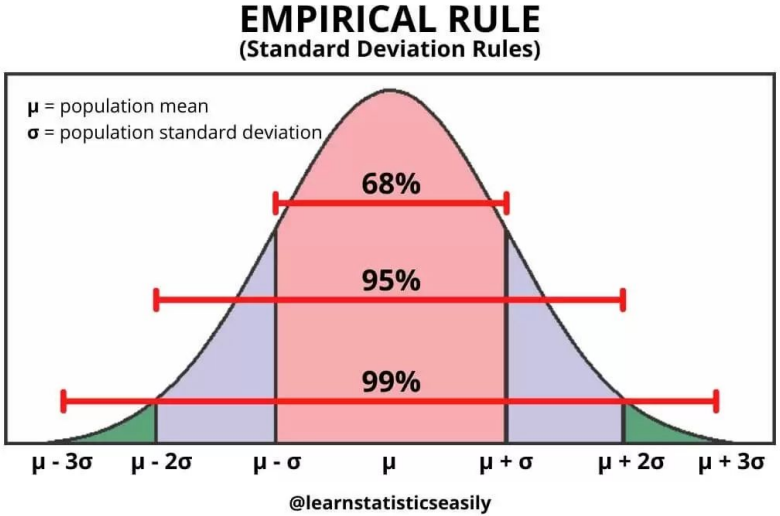
# Measurement of Spread

| Dataset A | Dataset B |
|---|---|
| 4, 5, 5, 5, 6, 6, 6, 6,7,7,7,8 | 1,2,3,4,5,6,6,7,8,9,10,11 |
| Mode:6, Mean: 6, Median: 6 | Mode:6, Mean: 6, Median: 6 |
| $x_i-\bar{x}$ = -2, -1, -1, -1, 0, 0, 0, 0, 1,1, 1, 2 | $x_i-\bar{x}$ = -5, -4, -3, -2, -1, 0, 0, 1, 2,3, 4, 5 |
| $(x_i-\bar{x})$^2 = 4, 1, 1, 1, 0, 0, 0, 0, 1,1, 1, 4 | $(x_i-\bar{x})$^2 = 25, 16, 9, 4, 1, 0, 0, 1, 4,9, 16, 25 |
| Variance = 1.17<br><br>Std. Deviation = 1.081 | Variance = 9.17<br><br>Std. Deviation = 3.03 |
| The larger variance and std, deviation in dataset B further demonstrate dataset b is more depressed than dataset A | |

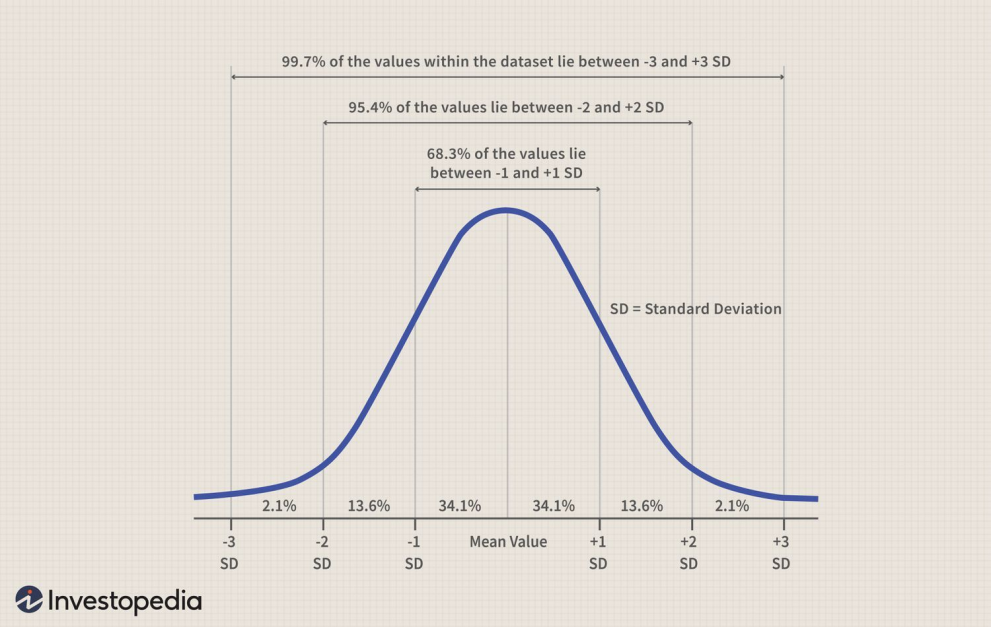
# Measurement of Spread

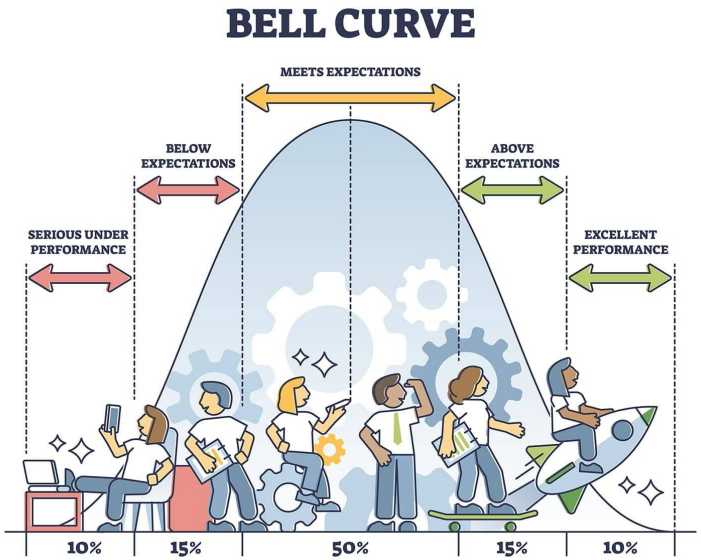


# Measurement of Spread

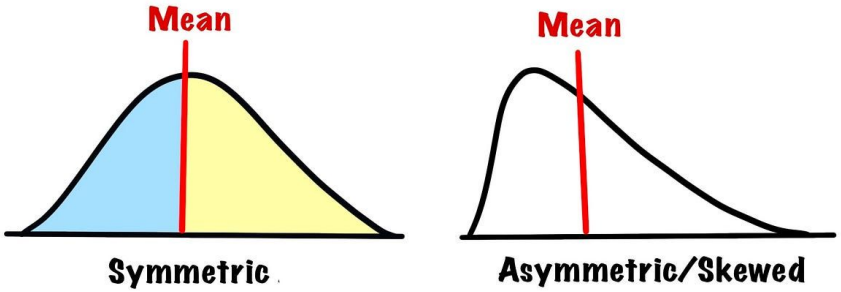


# Measure of shape

Measure of shape describe the distribution (or pattern) of the data within a dataset. The distribution shape of qualitative data can't be described as the data are not numeric. Broadly, it is divided into two types.

1. **Symmetrical**, the two sides of the distribution are mirror image of each other.
2. **Asymmetrical**, the two sides if the distribution are not mirror image of each other.



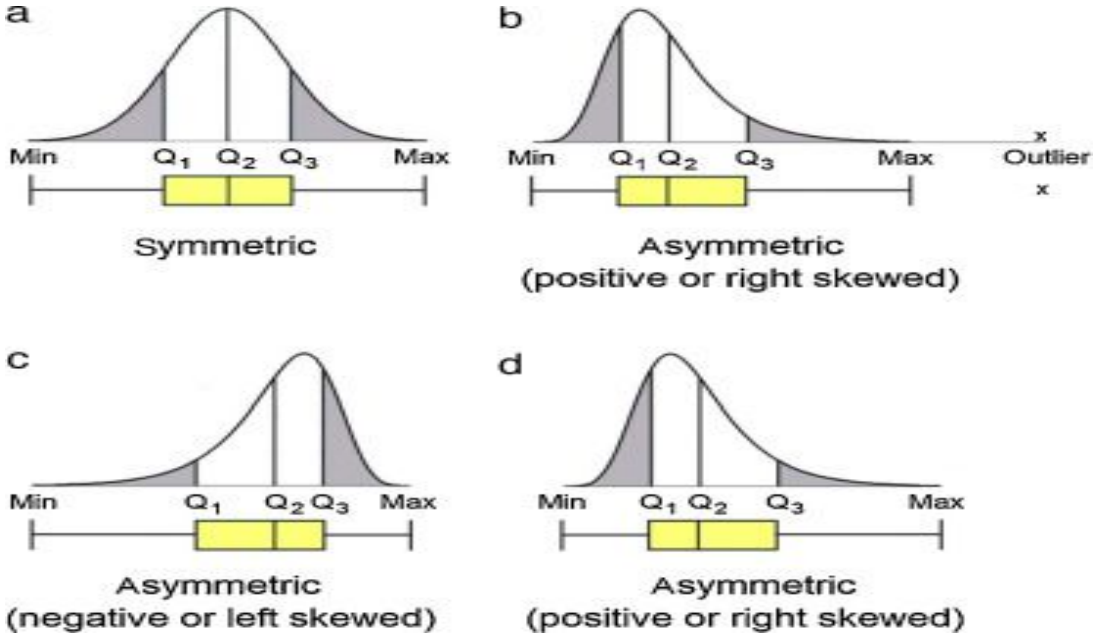
# Measure of shape

The shape of distribution can assist with identifying other descriptive statistics such as which central tendency is appropriate to use.

If the data are normally distributed, the mean, median and mode are equality and therefore all are appropriate measure of central tendency.

If the data are skewed the median may be more appropriate measure of central tendency.

# Measure of shape



# The effect of Transforming data on spread and centre

Suppose a Dataset of person weight

105, 156, 145, 172, 100

So Mean = 135.6 and Std. Deviation = 31.75

Suppose that during the winter, each person wear an extra five pound of clothes. Under these condition, What is the new mean and std. Deviation?

105+5, 156+5, 145+5, 172+5,100+5

110,161,150,177,105

Mean = 140.6 and Std Deviation = 31.75

# The effect of Transforming data on spread and centre

So, we can see that adding a constant to each data point transform the mean also by a constant but adding a constant doesn't change spread so there is no change in std. Deviation.

| Measure of central tendency | Measure of spread |
|---|---|
| Affected by : +, - , x, / | Affected by : x, / |
| Center_new = (Center_old)*A + B | Spread_new = (Spread_old)*A |

# The effects of outliers on spread and centre

| Measure | Outlier’s Effect |
|---|---|
| Mean | Affected |
| Median | Resistant |
| Mode | Resistant |
| Range | Affected |
| Std. Deviation | Affected |
| IQR | Resistant |

# Five number summary

Five number summary gives us a way to describe a distribution using only five number.

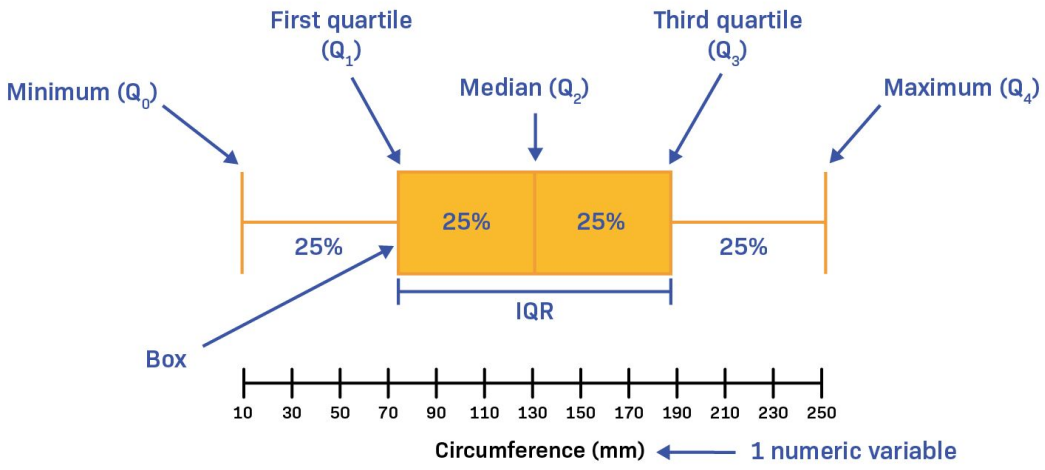
Data set

10, 11, 12, 25, 25, 27, 31, 33, 34, 34, 35, 36, 43, 50, 59

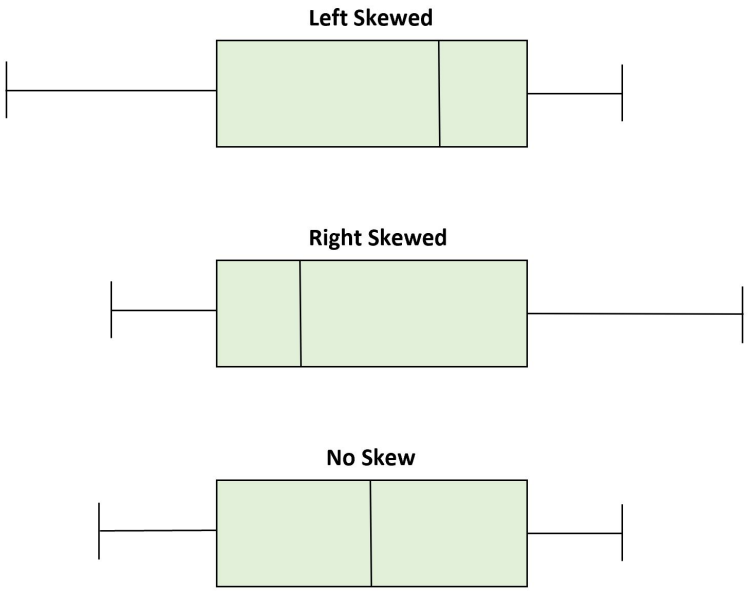
| Min | Q1 | Q2(median) | Q3 | Max |
|---|---|---|---|---|
| 10 | 25 | 33 | 36 | 59 |

# Five number summary

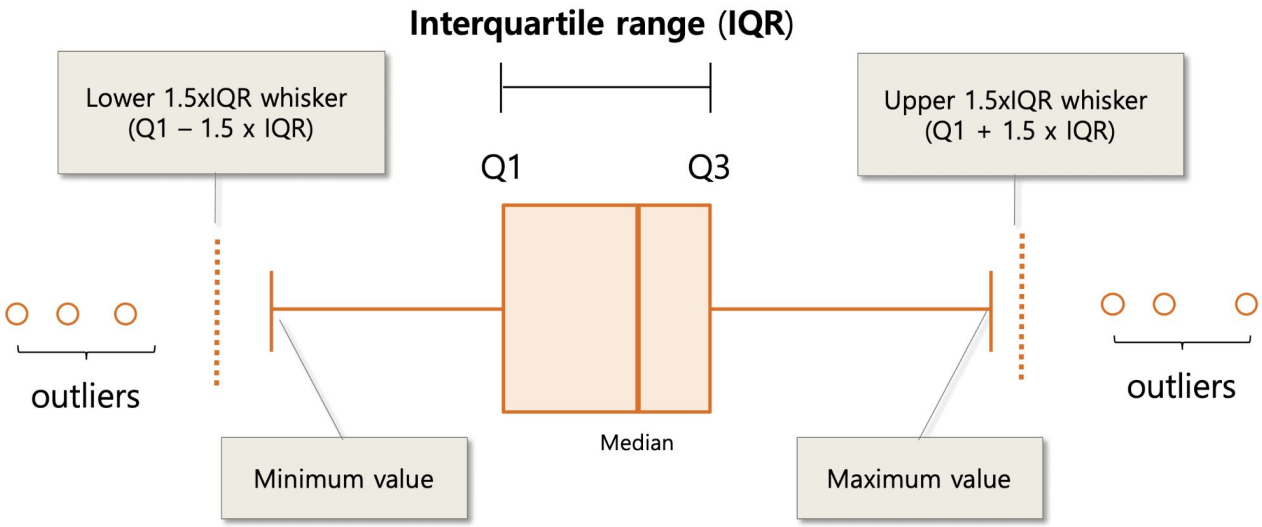
Using boxplot we can visualize the five number summary.



# Five number summary



# Five number summary

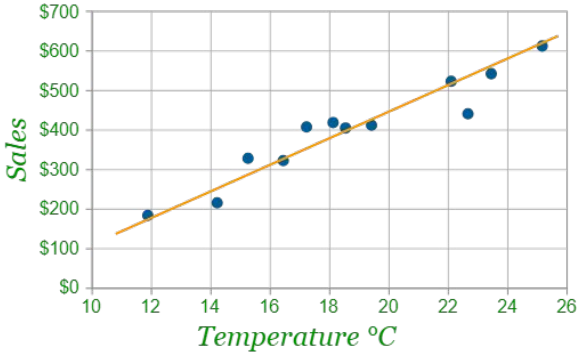


# Explanatory & Response Variable

**Explanatory variable** explain the outcome of a study. It is also called **independent variable**.

**Response variable** measure the outcome of a study. It is also called **dependent variable**.

**Time** plot and **Scatter** plot helps to describe the relationship between two variable.



# Covariance

**Covariance** is a measure of the relationship between two random variables and how much they change together.

Covariance can classify these three types of relationships.

1. Relationships with **positive** trends.
2. Relationships with **negative** trends.
3. **No relationships**

**Covariance**, in and of itself, is not very interesting. Instead covariance is a computational stepping stone to something that is interesting, like **correlation**.

# Covariance

Population Covariance Formula

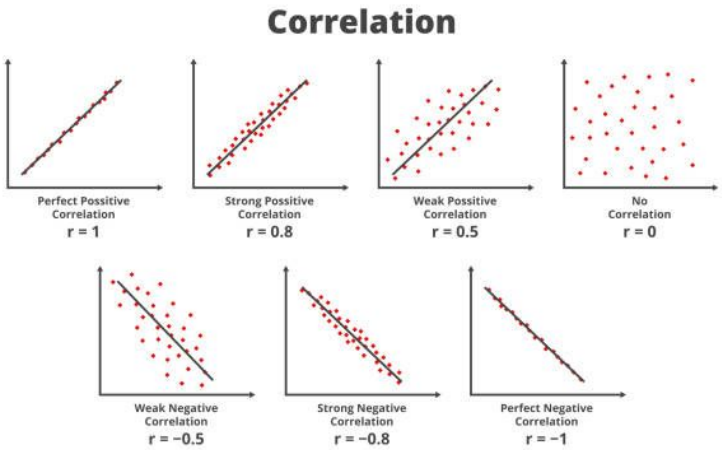
$$Cov(x,y) = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{N}$$

Sample Covariance

$$Cov(x,y) = \frac{\Sigma(x_i-\bar{x})(y_i-y)}{N-1}$$

# Correlation

Correlation tell us  about the direction and strength of a linear relationship shared between two quantitative variables.



# Correlation

**Sample correlation coefficient**

$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

$s_{xy}$ is the sample covariance

$s_x, s_y$ are the sample standard deviations.

**Population correlation coefficient**

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

$\sigma_{xy}$ is the population covariance

$\sigma_{x,}\sigma_y$ are the population standard deviations.

# Correlation

- r has value between 1 and -1.
- The strength of the linear relationship increased as r got close to 1 or -1.
- r = 0 means no association or correlation between the two variable.
- 0 < r < 0.25 weak correlation
- 0.25<= r < 0.75 intermediate correlation
- 0.75 <= r < 1 strong correlation
- r =1 perfect correlation

# Thanks